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14 July 2026

To the Editors of *Progress in Physics*
(Dr. Pierre Millette; Dr. Andreas Ries)

Dear Editors,

I am pleased to submit for your consideration the manuscript

"A Coulomb Route to Solar Fusion: $p + e + p \to d$ and $d + d \to {}^4$He"

for publication in *Progress in Physics* as a research article. In accordance with the journal's guidelines the manuscript is submitted as a preliminary PDF; a LaTeX version prepared in the `pip-template` will be provided upon acceptance.

What the paper does. It places two accounts of solar hydrogen burning side by side. In the standard proton–proton chain the first, rate-limiting step is the *weak* reaction $p + p \to d + e^+ + \nu_e$, which converts a proton into a neutron and emits a neutrino. In the RealNucleus picture — where the neutron is a bound proton–electron pair and the deuteron is $2p + 1e$ — the same step is purely *electromagnetic*: $p + e + p \to d$, a plasma electron captured as the deuteron's glue, with no flavour change, no positron, and no neutrino. The chain then closes with the directly-computed fusion $d + d \to {}^4$He. Both routes net to $4p + 2e^- \to {}^4$He at 26.73 MeV, differing only in the $\sim$2% the standard chain gives to neutrinos.

The central argument, and its honest limits. The clean distinction is one of lepton bookkeeping: in the Coulomb route no lepton is created or destroyed — the electron persists as the deuteron's constituent — so lepton-number conservation requires no neutrino, whereas the standard neutrino is the signature of the flavour-changing weak vertex $p \to n$. The paper is explicit about what this does and does not establish. It examines the original "desperate" need for the neutrino (the continuous beta spectrum and the spin–statistics of beta decay) and shows that the energy–momentum half lapses for a *capture* — $p + e + p \to d + \gamma$ emits the definite 2.22 MeV deuteron line — while the spin–statistics half does not, and is flagged as the open problem of the electron-in-nucleus picture. It concedes that the neutrino is indispensable to the Standard Model and rests on evidence broader than 1930, while noting that this evidence is theory-laden and that the neutrino's epistemic standing is nonetheless weaker than that of the electron and proton. The claim is therefore deliberately bounded: not that the neutrino is dispensable across physics, but that the entry step of solar burning may be electromagnetic rather than weak.

Why it is falsifiable. The two accounts are sharply opposed at one observable — the solar neutrino flux, locked to the luminosity at $\sim 2\,\nu$ per ^{4}He and directly measured (Borexino, SNO). The paper states this as the experimentum crucis rather than minimising it. The reactions $d + d \to {}^4$He and the deuteron and alpha structures are computed with an open-source, browser-based (WebGPU) RealQM solver, runnable from a public interactive gallery (`https://claes542.githu` b.io/RealMolecule/gallery.html; source https://github.com/Claes542/RealMolecule), so every computation is reproducible.

Declarations. The manuscript is original, is not under consideration elsewhere, and has not been previously published. There are no competing interests. The physical argument and the model are my own; the numerical implementation was developed and run with AI assistance (Claude, Anthropic), as stated in the Acknowledgements. A companion paper, *RealNucleus: Nuclear Binding as Dual Coulomb Confinement*, provides the nuclear computations on which this paper draws and is cited accordingly.

I would be glad to suggest reviewers or to provide any further material the editors may require. Thank you for your consideration.

Sincerely,

Claes Johnson
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